Ant colony optimization

Pseudo code

```
Ant colony optimization ()
initialize pheromone values on all paths
initialize parameter $\alpha, \beta, \rho$ & $Q$
while stopping conditions is not met DO
    for each ant DO
            Place ant at starting node
            while destination is not reached DO
                Select next node using:
                        pheromone level ($\tau$)
                        Heuristic information ($\eta$)
                Move to the selected node
        End while
        construct a complete solution
    End for
Evaluate all ant solutions
update pheromone trails
        Evaporate pheromones
        Deposit new pheromone based on
                solution quality
```

```
End whil
Return the best solution
```

Q2 Ans

Q:

Path planning

Title: Ant colony optimization for robot path planning

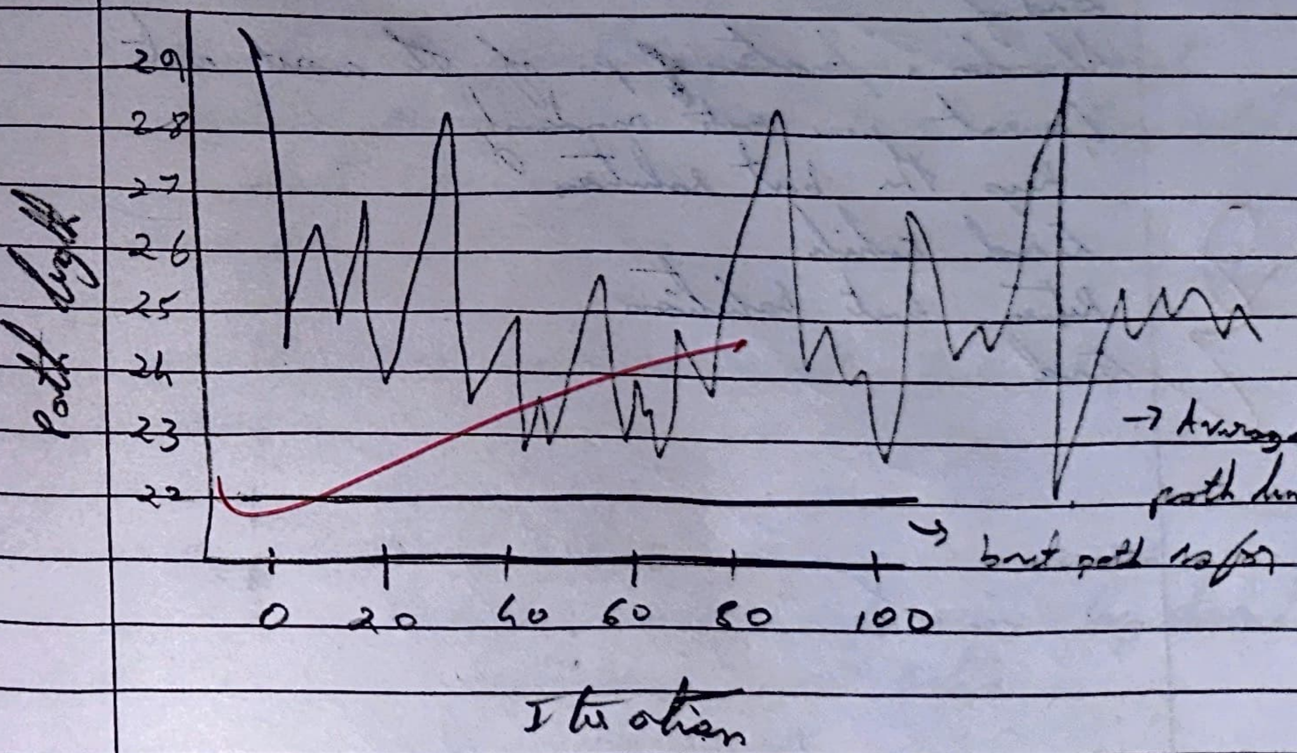
~~Ant~~ ACO is a nature inspired meta heuristic algorithm widely used to solve optimization problems. It is inspired by foraging behaviour of real ants which find efficient routes b/w their nests & food sources by depositing & following pheromones. In robot planning it is used to determine a feasible & efficient path from the robots initial position to a target destination while avoiding obstacles in environment

Code summary

The ACO-based robot path planning code initializes the environment, including the starting point, destination & obstacles & assigns initial pheromone values to possible paths. A group of artificial ants explore possible routes using pheromone intensity & heuristic information. The code evaluates the generated paths, updates pheromone levels through evaporation & ~~[illegible]~~ reinforcement, & repeats the process for a specified number of iteration or until a stopping condition is reached.

Output Summary

It displays the environment with the starting point, destination & obstacles & the best path found by the algorithm. It may also provide the total path length, number of iterations & the convergence behaviour. The resulting path is intended to avoid obstacles & minimize travel distance or another defined cost function. The quality of solution depends on the environment, algorithm parameters & stopping criteria



O/P verified
08/10/26.